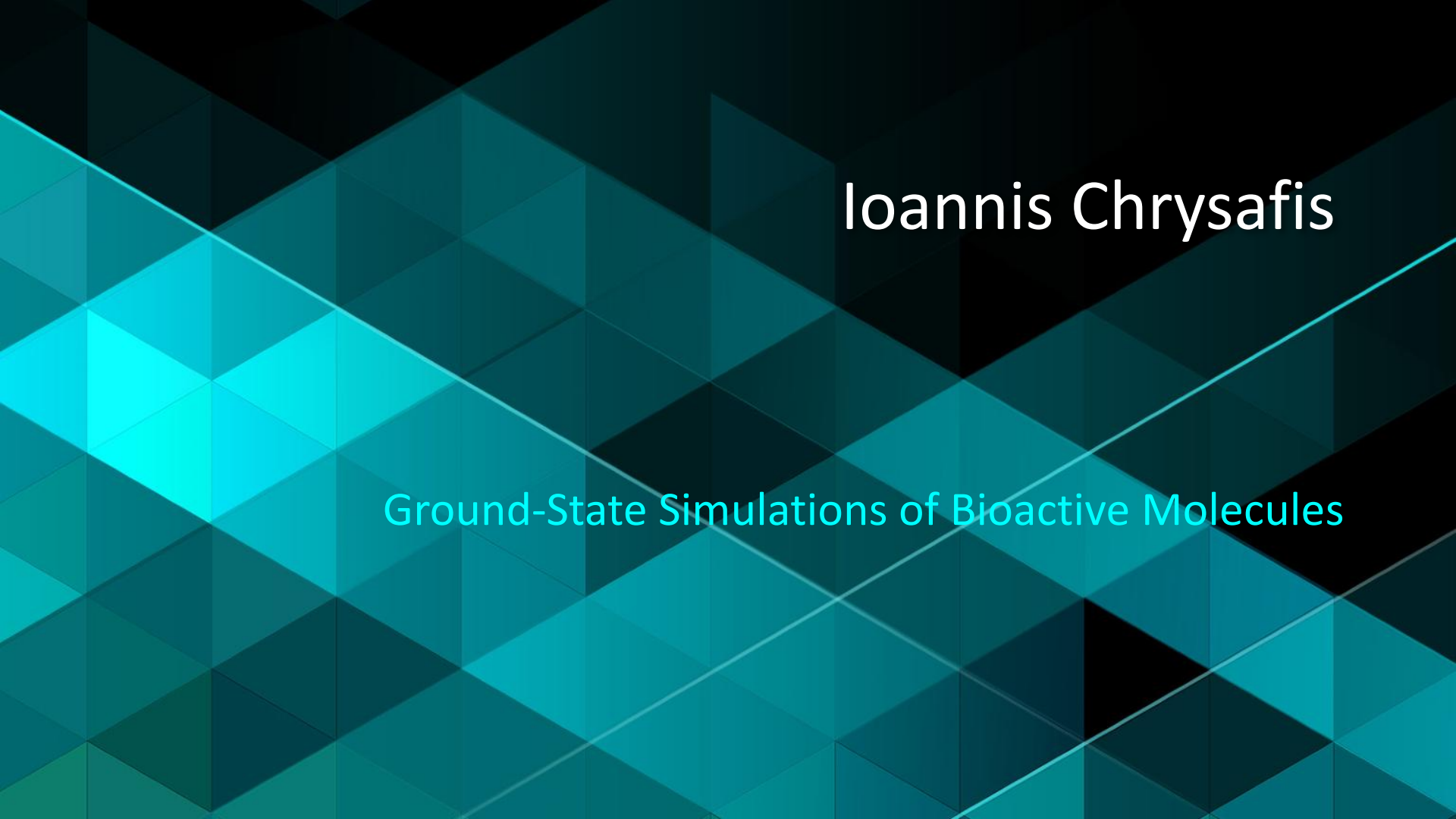




Ioannis Chrysafis

Ground-State Simulations of Bioactive Molecules



Contents

- Introducing CP2K
- Simulations
- Results

Introducing CP2K

CP2K is open-source software developed by programmers, chemists and physicists from several universities to simulate atomistic phenomena.

Introducing CP2K

It can run atomistic simulations of solid, liquid, molecular, periodic, material, crystal and biological systems.

CP2K provides a general framework for different modelling methods, such as:

Density Functional Theory (DFT)

using the mixed Gaussian and plane-wave approaches GPW and GAPW.

CP2K can perform molecular dynamics, metadynamics, Monte Carlo, Ehrenfest dynamics, vibrational analysis, core-level spectroscopy, energy minimisation and transition-state optimisation using the NEB or dimer method.

Simulations

In this thesis we searched for lower energy levels of several molecules by optimising their geometry.

Next, we present the code we used to simulate sildenafil and explain its main points.

CP2K Input Files

For the simulation we need to provide:

- Basis Sets
- Pseudopotentials
- System coordinates
 - Input file

CP2K Input Files

- Basis sets

Basis sets define the atomic orbitals for the electron density; here they are Gaussian functions:

```
# Hydrogen 6-31++G** (5s,1p) -> [3s,1p]
```

```
H 6-31++G**
```

```
4
```

```
1 0 0 3 1
```

```
0.1873113696E+02    0.3349460434E-01
```

```
0.2825394365E+01    0.2347269535E+00
```

```
0.6401216923E+00    0.8137573261E+00
```

```
1 0 0 1 1
```

```
0.1612777588E+00    1.00000000
```

```
1 0 0 1 1
```

```
0.36000000000E-01    0.1000000000E+01
```

```
1 1 1 1 1
```

```
0.11000000000E+01    1.00000000
```

It consists of 4 shells.

The first three are s-type ($l=0$) and the last one is p-type ($l=1$).

The last two shells describe polarisation and diffuse behaviour of the electron density.

CP2K Input Files

- Pseudopotentials

These are model potentials that we use to approximate the interaction of the electrons with the system. They are represented in the following form:

```
C GTH-PBE
```

```
1
```

```
6 2 2
```

```
0.2444330 0.0487790 0.0000000
```

```
0.0000000 0.0000000 0.0000000
```

```
0.5840000 0.0000000 0.0000000
```

```
1.0000000 0.0000000 0.0000000
```

```
1.0000000 1.0000000 0.0000000
```

It consists of the chemical symbol the pseudopotential is defined for, its type (GTH), the way electron interactions are treated (PBE), the atomic number (6) and the number of valence electrons (2s, 2p); the remaining lines give the pseudopotential's parameters.

CP2K Input Files

- System Coordinates

In this file we define the initial positions and types of the atoms.

48

```
C 4.983494 5.014580 2.631871
C -0.212473 0.729220 4.588228
C 4.389003 3.774356 1.987043
C 0.382018 1.969444 5.233055
C 0.199817 1.745745 1.103809
C 4.571204 3.998055 6.116290
H 4.200651 5.453753 2.765298
H 0.570370 0.290046 4.454801
H 0.504381 4.693253 3.610049
H 4.266640 1.050547 3.610049
H 5.417761 5.718424 2.043288
H -0.646740 0.025376 5.176811
H 3.755155 3.318266 2.584795
H 1.015865 2.425533 4.635304
H 3.907823 3.892137 1.025254
H 0.863198 1.851662 6.194845
H 2.153340 1.184326 1.205756
H 2.617681 4.559474 6.014342
H 1.188951 0.152902 0.483747
H 3.582070 5.590898 6.736352
N 1.293886 0.996374 0.926194
N 3.477135 4.747425 6.293905
O 0.485510 2.898964 1.717300
O 4.205511 2.844025 5.502700
```

It consists of the number of atoms,
their types and their coordinates.

CP2K Input Files

```
&GLOBAL
  PROJECT sildenafil_energy
  RUN_TYPE GEO_OPT
  PRINT_LEVEL LOW
  WALLTIME 172520
&END GLOBAL

&FORCE_EVAL
  METHOD Quickstep
&DFT
  BASIS_SET_FILE_NAME EMSL_BASIS_SETS
  POTENTIAL_FILE_NAME GTH_POTENTIALS_PSI
  CHARGE 0

  &MGRID
    CUTOFF 300
    REL_CUTOFF 50
  &END MGRID

  &QS
    METHOD GAPW

    EXTRAPOLATION ASPC
    EXTRAPOLATION_ORDER 3
  &END QS

  &SCF
    MAX_SCF 200
    SCF_GUESS ATOMIC
    EPS_SCF 1.0E-6
    CHOLESKY INVERSE
    &OT ON
      MINIMIZER CG
      PRECONDITIONER FULL_KINETIC
      ENERGY_GAP 0.001
    &END OT

  &OUTER_SCF
    EPS_SCF 1.0E-4
    MAX_SCF 50
  &END OUTER_SCF
&END SCF

  &XC
    &XC_FUNCTIONAL PBE
  &END XC_FUNCTIONAL
  &END XC
&END DFT
```

The main sections of the input are:

- Global, which sets the type of simulation
- Force Eval, which holds the parameters for computing the system's energy.
- Motion, which holds the parameters for geometry optimisation.

CP2K Input Files

```
&GLOBAL
PROJECT sildenafil_energy
RUN_TYPE GEO_OPT
PRINT_LEVEL LOW
WALLTIME 172520
&END GLOBAL

&FORCE_EVAL
METHOD Quickstep
&DFT
BASIS_SET_FILE_NAME EMSL_BASIS_SETS
POTENTIAL_FILE_NAME GTH_POTENTIALS_PSI
CHARGE 0

&MGRID
CUTOFF 300
REL_CUTOFF 50
&END MGRID

&QS
METHOD GAPW

EXTRAPOLATION ASPC
EXTRAPOLATION_ORDER 3
&END QS

&SCF
MAX_SCF 200
SCF_GUESS ATOMIC
EPS_SCF 1.0E-6
CHOLESKY INVERSE
&OT ON
MINIMIZER CG
PRECONDITIONER FULL_KINETIC
ENERGY_GAP 0.001
&END OT

&OUTER_SCF
EPS_SCF 1.0E-4
MAX_SCF 50
&END OUTER_SCF
&END SCF

&XC
&XC_FUNCTIONAL PBE
&END XC_FUNCTIONAL
&END XC
&END DFT
```

The electronic structure is computed with the Quickstep method according to Density Functional Theory (DFT)

In the DFT section we declare the parameters related to Density Functional Theory, such as the basis sets and pseudopotentials mentioned above, as well as properties of the grid on which the electronic equations are expanded (MGRID), such as the distribution of its points and their interaction.

CP2K Input Files

```
&GLOBAL
  PROJECT sildenafil_energy
  RUN_TYPE GEO_OPT
  PRINT_LEVEL LOW
  WALLTIME 172520
&END GLOBAL

&FORCE_EVAL
  METHOD Quickstep
  &DFT
    BASIS_SET_FILE_NAME EMSL_BASIS_SETS
    POTENTIAL_FILE_NAME GTH_POTENTIALS_PSI
    CHARGE 0

    &MGRID
      CUTOFF 300
      REL_CUTOFF 50
    &END MGRID

    &QS
      METHOD GAPW

      EXTRAPOLATION ASPC
      EXTRAPOLATION_ORDER 3
    &END QS

    &SCF
      MAX_SCF 200
      SCF_GUESS ATOMIC
      EPS_SCF 1.0E-6
      CHOLESKY INVERSE
      &OT ON
        MINIMIZER CG
        PRECONDITIONER FULL_KINETIC
        ENERGY_GAP 0.001
      &END OT

    &OUTER_SCF
      EPS_SCF 1.0E-4
      MAX_SCF 50
    &END OUTER_SCF
  &END SCF

  &XC
    &XC_FUNCTIONAL PBE
  &END XC_FUNCTIONAL
  &END XC
&END DFT
```

In the QS section we declare the method used to represent and compute the electron potentials.

In the SCF (Self-Consistent Field) section these potentials are parameterised.

CP2K Input Files

```
&GLOBAL
  PROJECT sidenafil_energy
  RUN_TYPE GEO_OPT
  PRINT_LEVEL LOW
  WALLTIME 172520
&END GLOBAL

&FORCE_EVAL
  METHOD Quickstep
  &DFT
    BASIS_SET_FILE_NAME EMSL_BASIS_SETS
    POTENTIAL_FILE_NAME GTH_POTENTIALS_PSI
    CHARGE 0

    &MGRID
      CUTOFF 300
      REL_CUTOFF 50
    &END MGRID

    &QS
      METHOD GAPW

      EXTRAPOLATION ASPC
      EXTRAPOLATION_ORDER 3
    &END QS

    &SCF
      MAX_SCF 200
      SCF_GUESS ATOMIC
      EPS_SCF 1.0E-6
      CHOLESKY INVERSE
      &OT ON
        MINIMIZER CG
        PRECONDITIONER FULL_KINETIC
        ENERGY_GAP 0.001
      &END OT

    &OUTER_SCF
      EPS_SCF 1.0E-4
      MAX_SCF 50
    &END OUTER_SCF
  &END SCF

  &XC
    &XC_FUNCTIONAL PBE
  &END XC_FUNCTIONAL
  &END XC
&END DFT
```

This parameterisation follows these steps:

- Initial guess of the wavefunction (SCF_GUESS)
- Construction of the potentials
- Solving the Schrödinger equations to build the potentials
- Computing the electron density from the new wavefunctions
- Comparison with the previous density (return to a new guess, or continue)
- Mixing of the old and new density
- Convergence check (EPS_SCF)
- If converged, we move on to the next section; otherwise the procedure repeats up to MAX_SCF times.

CP2K Input Files

```
&GLOBAL
PROJECT sildenafil_energy
RUN_TYPE GEO_OPT
PRINT_LEVEL LOW
WALLTIME 172520
&END GLOBAL

&FORCE_EVAL
METHOD Quickstep
&DFT
BASIS_SET_FILE_NAME EMSL_BASIS_SETS
POTENTIAL_FILE_NAME GTH_POTENTIALS_PSI
CHARGE 0

&MGRID
CUTOFF 300
REL_CUTOFF 50
&END MGRID

&QS
METHOD GAPW

EXTRAPOLATION ASPC
EXTRAPOLATION_ORDER 3
&END QS

&SCF
MAX_SCF 200
SCF_GUESS ATOMIC
EPS_SCF 1.0E-6
CHOLESKY INVERSE
&OT ON
MINIMIZER CG
PRECONDITIONER FULL_KINETIC
ENERGY_GAP 0.001
&END OT

&OUTER_SCF
EPS_SCF 1.0E-4
MAX_SCF 50
&END OUTER_SCF
&END SCF

&XC
&XC_FUNCTIONAL PBE
&END XC_FUNCTIONAL
&END XC
&END DFT
```

In this simulation we chose to carry out the procedure above with the orbital transformation (OT) method: we transformed the basis sets, the orbitals and the Hamiltonian matrix and followed the same steps. The main differences are that, at each iteration without convergence, the orbital transformation is updated (CG) with a preconditioner; that the resulting matrices are sparser, because the molecular orbitals have significant values only around the atoms; and that the basis sets and most calculations use orthogonalised matrices (CHOLESKY INVERSE).

CP2K Input Files

```
&GLOBAL
  PROJECT sildenafil_energy
  RUN_TYPE GEO_OPT
  PRINT_LEVEL LOW
  WALLTIME 172520
&END GLOBAL

&FORCE_EVAL
  METHOD Quickstep
&DFT
  BASIS_SET_FILE_NAME EMSL_BASIS_SETS
  POTENTIAL_FILE_NAME GTH_POTENTIALS_PSI
  CHARGE 0

  &MGRID
    CUTOFF 300
    REL_CUTOFF 50
  &END MGRID

  &QS
    METHOD GAPW

    EXTRAPOLATION ASPC
    EXTRAPOLATION_ORDER 3
  &END QS

  &SCF
    MAX_SCF 200
    SCF_GUESS ATOMIC
    EPS_SCF 1.0E-6
    CHOLESKY INVERSE
    &OT ON
      MINIMIZER CG
      PRECONDITIONER FULL_KINETIC
      ENERGY_GAP 0.001
    &END OT

  &OUTER_SCF
    EPS_SCF 1.0E-4
    MAX_SCF 50
  &END OUTER_SCF
&END SCF

&XC
  &XC_FUNCTIONAL PBE
&END XC_FUNCTIONAL
&END XC
&END DFT
```

The OUTER_SCF section sets how many times the previous procedure is repeated: if it reaches a new stable state with lower total energy, the procedure repeats until the OUTER_SCF conditions are met.

The XC section defines the electron–electron interactions not included in the Hamiltonian (XC_FUNCTIONAL PBE).

Introducing CP2K

```
&END DFT

&SUBSYS
  &CELL
    A 19.000000000 0.000000000 0.000000000
    B 0.000000000 17.000000000 0.000000000
    C -6.000000000 0.000000000 45.000000000
    PERIODIC XYZ
  &END CELL

  &TOPOLOGY
    COORD_FILE_NAME sidenafil.xyz
    COORD_FILE_FORMAT XYZ
  &END

  &KIND H
    BASIS_SET 6-31++Gxx
    POTENTIAL GTH-PBE-q1
  &END KIND

  &KIND C
    BASIS_SET 6-31++Gxx
    POTENTIAL GTH-PBE-q4
  &END KIND

  &KIND N
    BASIS_SET 6-31++Gxx
    POTENTIAL GTH-PBE-q5
  &END KIND

  &KIND O
    BASIS_SET 6-31++Gxx
    POTENTIAL GTH-PBE-q6
  &END KIND

  &KIND S
    BASIS_SET 6-31++Gxx
    POTENTIAL GTH-PBE-q6
  &END KIND
&END SUBSYS
```

The SUBSYS section defines the dimensions of the simulation cell (CELL), the initial arrangement of the atoms (TOPOLOGY), and the pseudopotentials and basis sets supplied by the user.

Introducing CP2K

```
&END SUBSYS
```

```
&PROPERTIES
```

```
  &RESP
```

```
    &SPHERE_SAMPLING
```

```
      AUTO_VDW_RADII_TABLE CAMBRIDGE
```

```
      AUTO_RMIN_SCALE 1.0
```

```
      AUTO_RMAX_SCALE 100.0
```

```
    &END
```

```
    &PRINT
```

```
      &RESP_CHARGES_TO_FILE
```

```
      &END RESP_CHARGES_TO_FILE
```

```
      &PROGRAM_RUN_INFO
```

```
      &END
```

```
      &COORD_FIT_POINTS
```

```
      &END
```

```
    &END PRINT
```

```
  &END RESP
```

```
  &FIT_CHARGE
```

```
  &END FIT_CHARGE
```

```
&END PROPERTIES
```

```
&END FORCE_EVAL
```

The PROPERTIES section defines the properties and quantities we want to compute from the simulation results, such as the atomic arrangement obtained after each SCF convergence (COORD_FIT_POINTS).

Introducing CP2K

```
&END FORCE_EVAL
```

```
&MOTION
```

```
  &CELL_OPT
```

```
    OPTIMIZER LBFGS
```

```
      TYPE DIRECT_CELL_OPT
```

```
      &PRINT
```

```
        &CELL
```

```
        &END
```

```
    &END PRINT
```

```
  &END CELL_OPT
```

```
&END MOTION
```

The MOTION section defines the parameters for molecular dynamics and geometry optimisation (OPTIMIZER). We also chose to optimise the geometry and compute the electronic structure simultaneously (DIRECT_CELL_OPT).

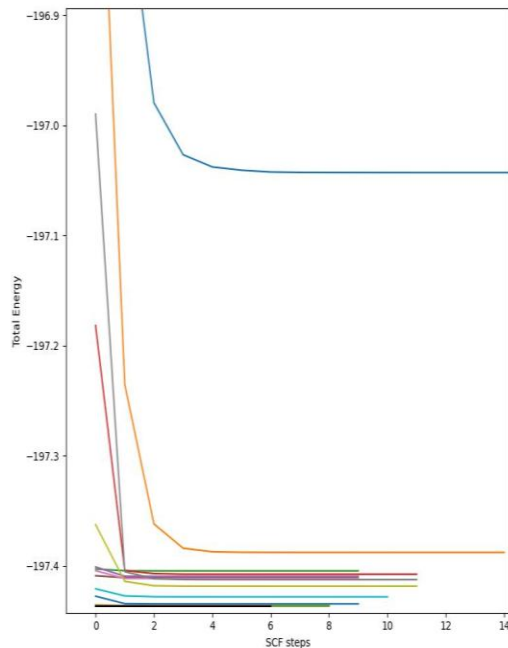
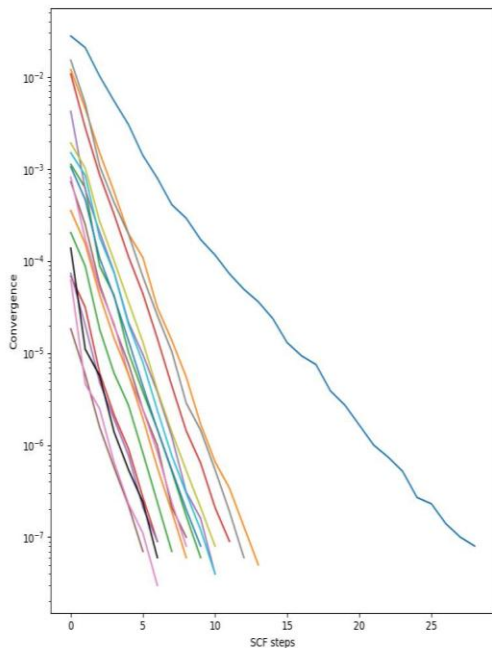
Simulations

In this thesis we ran geometry-optimisation simulations for paracetamol, sildenafil, cytosine, cyclohexanedione, ethyl carbamate, oxalic acid and formamide.

Checking the results

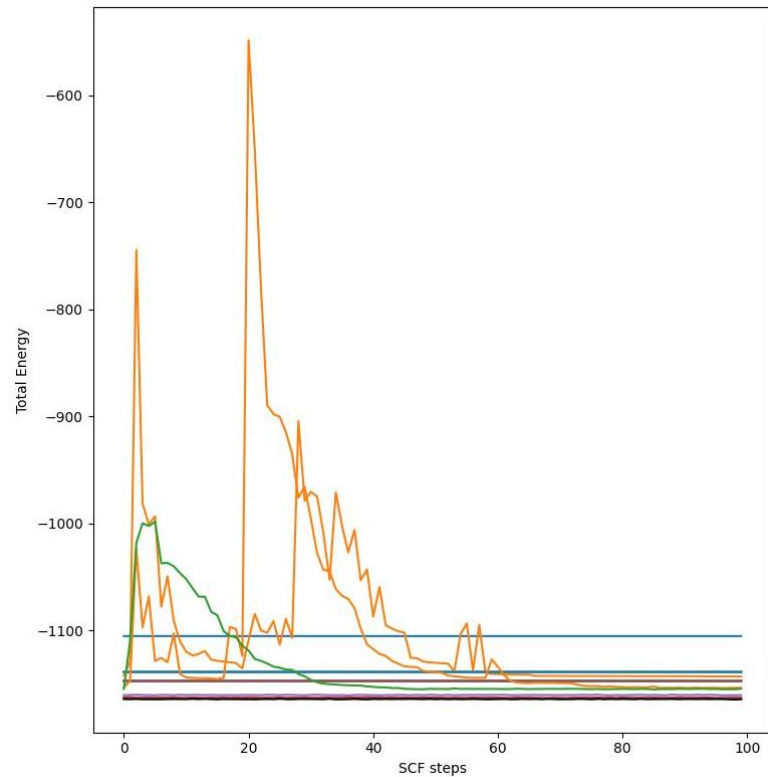
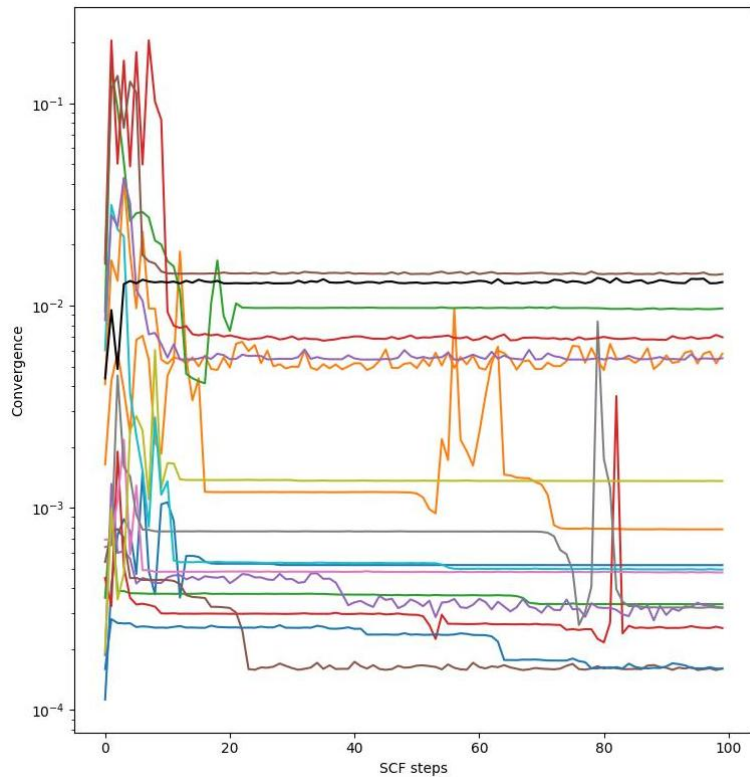
For all the simulations above we followed this procedure:
First, by simulating the energy of a single point, we chose the best combination of CUTOFF and REL_CUTOFF — that is, the distance beyond which interactions with other grid points are negligible, and the grid density. We made this choice by comparing the run time for each combination.

Checking the results



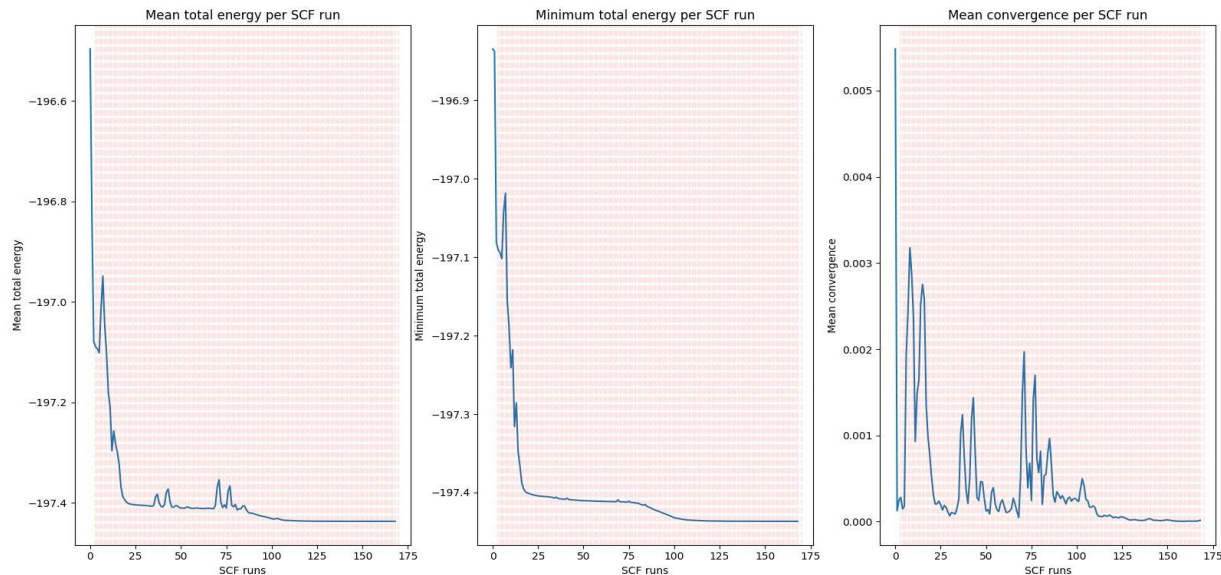
Next we studied the minimum energy and the convergence at each SCF step, as shown in the plot. In this way we chose a suitable number of iterations (MAX_SCF) and checked whether the parameterisations converge to a single wavefunction.

Checking the results



Checking the results

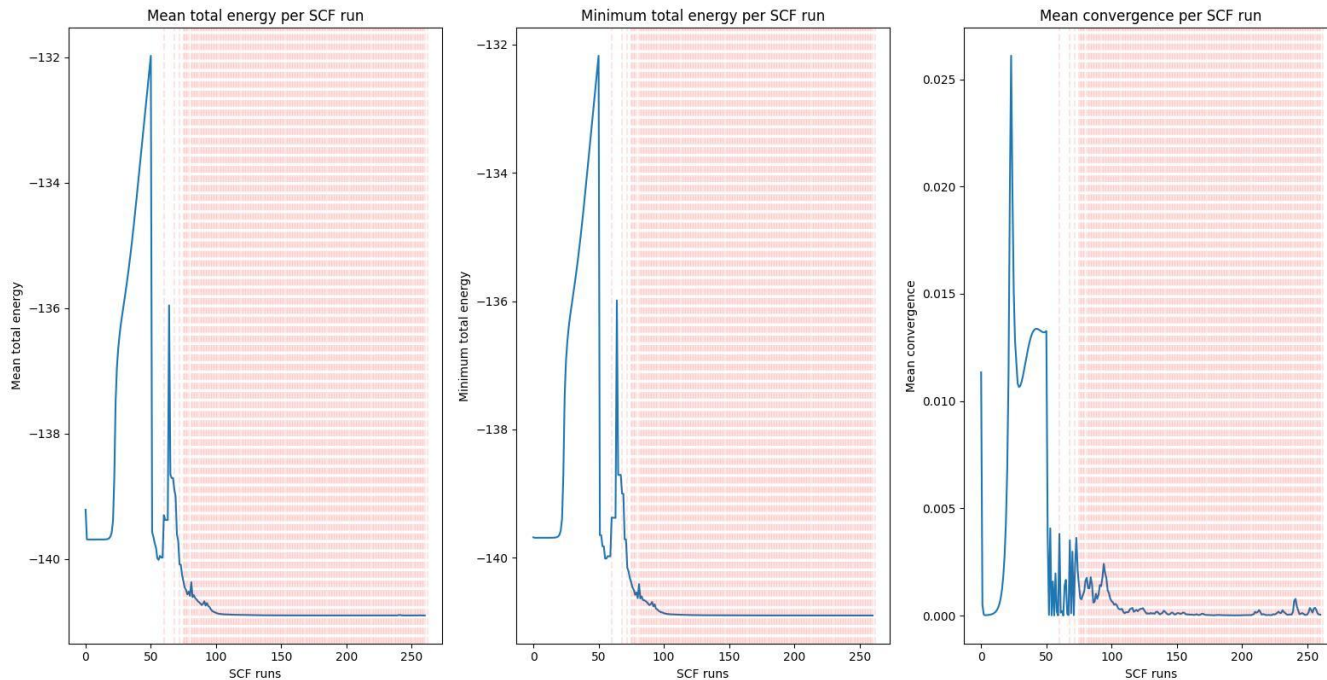
mean total energy: -197.38703467772444 minimum total energy: -197.4367022592 mean convergence: 0.00043301522943890914



- After each SCF run we also studied the mean energy, the minimum energy and the mean convergence, to check whether the simulation leads to a more stable state with lower minimum energy and whether the number of iterations is sufficient.

Checking the results

mean total energy: -139.95001033934415 minimum total energy: -140.898734699 mean convergence: 0.0019940536833025634



Handling errors

The main problem we faced was that the simulations did not converge to a stable state, or converged very slowly.

This was due to one of the following:

- An unsuitable method for computing the electron potentials
 - An unsuitable method for the orbital transformation
- An unsuitable method for optimising the molecular geometry

Handling errors

We resolved these errors experimentally, checking the results after every change to the simulation.